

Thesis Proposal

Title: Multi-Agent system for automated mining accident investigation using large language models and argumentation based reasoning

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1. Problem statement and motivation

Mining accidents remain a critical safety concern worldwide, with investigations requiring analysis of diverse evidence sources: sensor logs, equipment maintenance records, witness statements, regulatory documents, and environmental data. Current investigation processes are labor-intensive, often taking months to complete, and heavily dependent on expert availability. The complexity of synthesizing heterogeneous data sources while maintaining traceable reasoning chains presents a significant challenge.

Large Language models offer promising capabilities for processing unstructured text and reasoning over evidence. However, when multiple analytical perspectives yield conflicting conclusions, existing approaches lack principled mechanisms for conflict resolution and explanation generation. Investigators need not just conclusions but verifiable justifications that can withstand legal and regulatory scrutiny.

This thesis addresses the gap between AI-assisted evidence analysis and the explainability requirements of formal investigation processes.

2. Research Objectives

The primary objective is to design and implement a multi-agent that assists investigators in analyzing mining accident evidence and generating structured, explainable investigation reports. Specific Objectives:

1. Evidence Processing Pipeline: Develop a system that ingests heterogeneous data sources (sensor logs, maintenance records, textual reports, regulatory standards) and creates a unified knowledge representation suitable for agent reasoning.
2. Specialist Agent Architecture: Design agents that analyse evidence from distinct perspectives (equipment failure analysis, human factors, environmental conditions, regulatory compliance), each producing structured claims with supporting evidence.
3. Argumentation-based conflict resolution: Implement a formal argumentation layer that resolves conflicts between agent findings using attack/support relations and computes acceptable conclusions based on Dung's argumentation semantics.
4. Explainable Report generation: Generate investigation reports that include not only conclusions but the full argumentation graph showing how evidence supports or contradicts each finding.

3. Methodology

Phase 1: Literature Review and Framework Design

- Survey existing approaches to automated incident investigation

- Study argumentation theory (Dung's abstract argumentation, ASPIC+) and its application to evidence-based reasoning
- Design the system architecture integrating RAG, multi-agent coordination, and argumentation semantics

Phase 2: Data Collection and Preprocessing

- Collect publicly available mining accident reports(e.g MSHA database, Rostekhnadzor reports)
- Develop preprocessing pipelines for structured and unstructured data
- Create knowledge base of mining safety regulations and equipment specifications

Phase 3: System Implementation

- Implement specialist agents using LangChain/LangGraph with tool-calling capabilities
- Develop the argumentation layer for conflict detection and resolution
- Build the report generation module with justification traces

Phase 4: Evaluation and Validation

- Evaluate system against historical accidents reports with known conclusions
- Conduct qualitative assessment with domain experts(if available)
- Measure explanation quality and reasoning coherence

Phase 5: Thesis Writing

- Document methodology, implementation and results
- Prepare defense presentation

4. Technical Approach (subjected to changes)

Agent Architecture:

- Each specialist agent uses RAG to access relevant knowledge (equipment manuals, safety standards, similar past incidents)
- Agents produce structured outputs: {claim, evidence, confidence, warrant}
- Tool-calling enables agents to query databases, perform calculations, and retrieve specific documents

Argumentation Layer:

- Claims from different agents are mapped to an argumentation framework
- Attack relations are identified when claims contradict (e.g., "equipment failure caused accident" vs. "human error caused accident")
- Grounded semantics or preferred semantics compute the set of acceptable arguments
- The final report includes the argumentation graph as a justification artifact

Technology Stack:

- Python, LangChain/LangGraph for agent orchestration
- ChromaDB/FAISS for vector storage
- Open-source LLMs (LLaMA, Mistral) or API-based models (GPT-4, Claude)
- NetworkX for argumentation graph computation

5. Expected Contributions

1. **Practical System:** A working prototype that demonstrates AI-assisted accident investigation with explainable outputs
2. **Methodological Framework:** A generalizable approach for applying argumentation-based multi-agent reasoning to evidence analysis tasks
3. **Evaluation Insights:** Empirical findings on the effectiveness of argumentation-based explanations for investigation tasks